

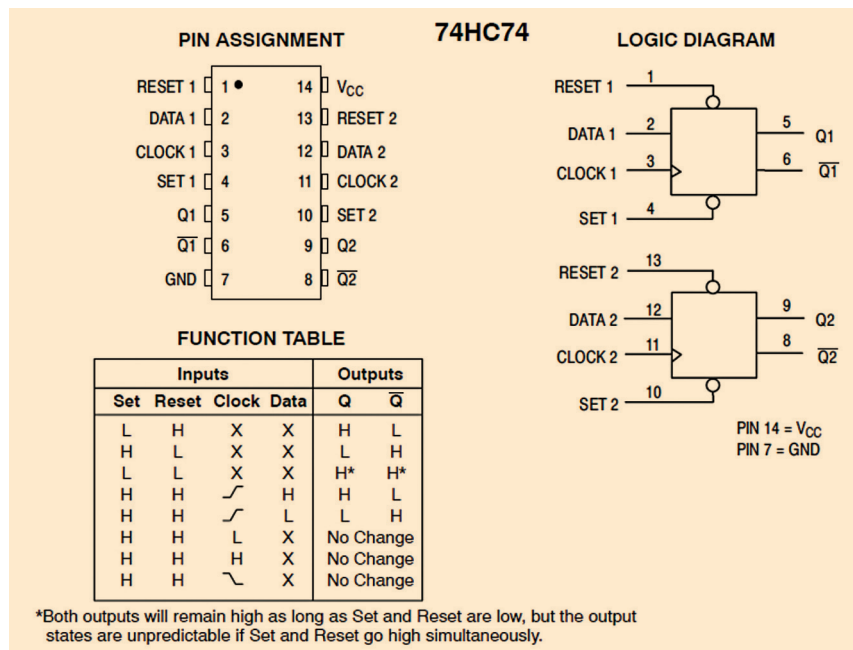


Fig. 4: Pin diagram and truth table of 74HC74 (Credit: ON Semiconductor)

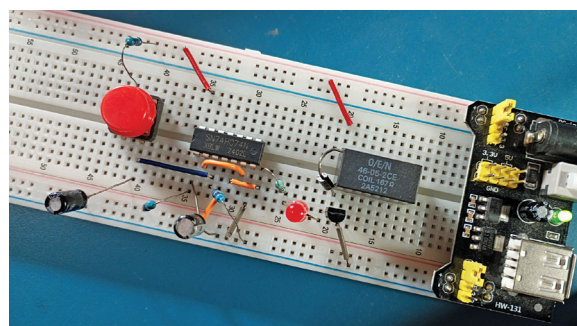


Fig. 5: Author's prototype on breadboard

inputs. The Set and Reset inputs operate asynchronously.

The logic state at the D input is transferred to the corresponding Q output on the rising edge of the clock input. Both Q and $\bar{Q}$ (inverted Q) outputs are available from each flip-flop. Fig. 4 presents a relevant excerpt from the device datasheet.

As shown in Fig. 3, the operation is straightforward. The push-button switch S1 provides the clock signal to the D flip-flop. The $\bar{Q}1$ output (Pin 6) is fed back to the D1 input (Pin 2), enabling the Q1 output (Pin 5) to toggle with each pulse on the CLOCK1 input (Pin 3).

Construction and testing

After assembling the circuit on a breadboard, briefly press switch S1. LED1 lights up, indicating relay energisation and load activation. Press S1 again to de-energise the relay and disable the load.

Fig. 5 illustrates the prototype

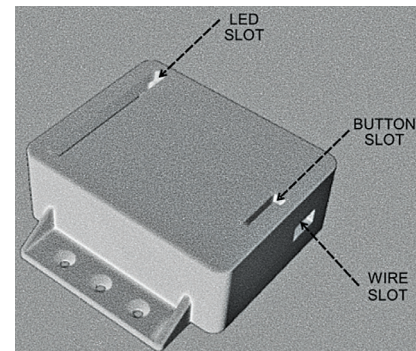


Fig. 6: Proposed enclosure for the relay module

PARTS LIST

Semiconductors:

- IC1 (IC1A, IC1B, IC1C) - 74HC74 D-type flip-flop
- T1 - 2SC1815 NPN transistor
- D1 - 1N4007 rectifier diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2 - 10-kilo-ohm
- R3 - 100-kilo-ohm
- R4 - 1-kilo-ohm

Capacitors:

- C1, C2 - 1 μ F, 16V electrolytic
- C3 - 100n ceramic disk

Miscellaneous:

- CON1 - 2-pin connector
- RL1 - 5V SPDT relay
- S1 - Push-to-on switch
- 5V power supply

assembled on a breadboard. For practical use, enclosing the system in a compact case is recommended. Fig. 6 provides a sample enclosure layout.

Note. As the 74HC74 comprises two D flip-flops, it can be used to construct two independent toggle relay segments within a single system. **EFY**



T.K. Hareendran is an electronics designer, hardware beta tester, technical author, and product reviewer

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